INCH-POUND

MS33781B
9 JULY 1996
SUPERSEDING
MS33781A
24 JUNE 1965

MILITARY SPECIFICATION SHEET
RECESS, OFFSET CRUCIFORM,
DIMENSIONS OF RECESS, GAGE AND DRIVER FOR

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specification and Standards (DODISS) specified in the solicitation: NONE

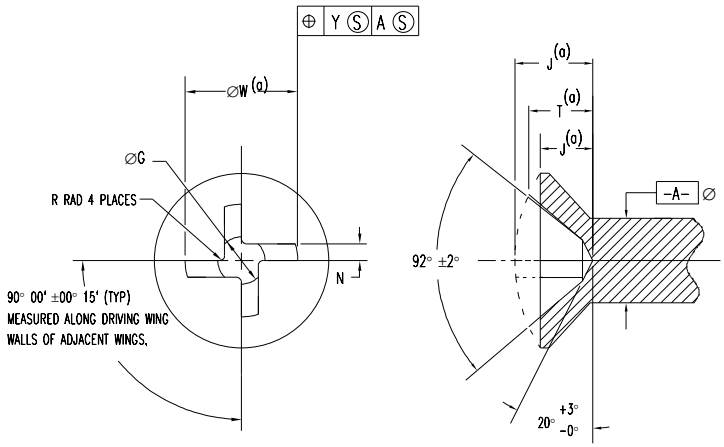


FIGURE 1. OFFSET CRUCIFORM RECESS[b]

Table I. Offset Cruciform Recess Dimensions [c]

Recess Size	N		∅G		R		Y
	Max	Min	Max	Min	Max	Min	Tol
0	.018	.016	.037	.033	.010	.000	.012
1			.044	.040			
2			.052	.048			
3	.021	.018	.059	.055			
4	.023	.020	.067	.063	.018	.008	
5	.026	.023	.075	.071			
6	.029	.026	.082	.078	.030	.016	
8	.034	.031	.096	.092			
10	.039	.036	.111	.107	.040	.023	
1/4	.051	.048	.147	.143			.016
5/16	.064	.061	.184	.180	.052	.031	
3/8	.076	.073	.220	.216			
7/16	.089	.086	.256	.252	.072	.048	
1/2	.102	.098	.292	.288	.089	.062	
9/16	.115	.111	.328	.324	.108	.078	.024
5/8	.127	.123	.365	.361			

(a) The specific part standard or drawing which specifies this recess drive referencing this document shall specify values for dimensions J, W, T, and depth GAGE PENETRATION, if the values specified herein are not compatible with the part configuration or performance requirements.

(b) These recesses are also produced and marketed by the Licensees of Phillips Screw Company under the designation of Torq-Set®.

(c) For dimensions of ribbed recesses, sizes 6 through 5/16 inch, see MS14191.

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T17:08Z
Check the source to verify that this is the current version before use.

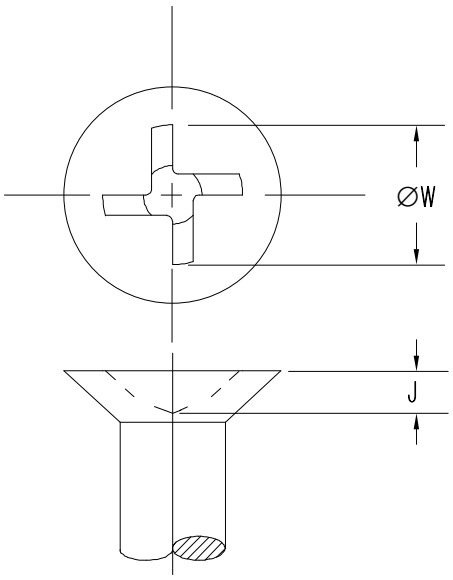


FIGURE 2

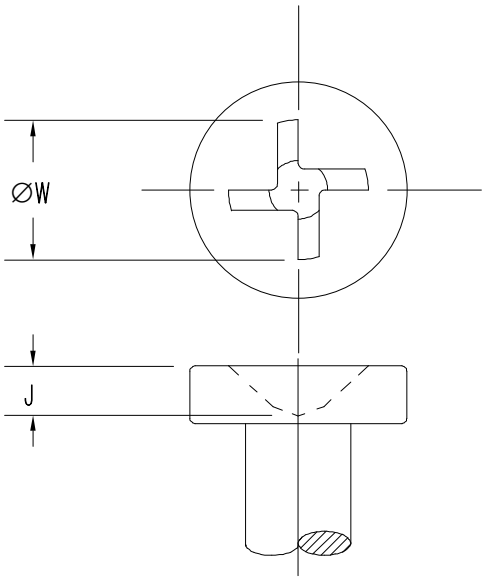


FIGURE 3

Table II [(a)]

100° Flush Head

Screw Size	Recess and Driver Size	∅W Wing	J Recess Depth	GAGE PENET
0	0	.082 .072	.031 .021	.0225 .0145
1	1	.098 .088	.037 .027	.0270 .0185
2	2	.115 .105	.043 .033	.0315 .0225
3	3	.132 .122	.049 .039	.0365 .0270
4	4	.148 .138	.055 .045	.0405 .0305
5	5	.165 .155	.061 .051	.0450 .0350
6	6	.182 .172	.066 .056	.0500 .0395
8	8	.215 .205	.078 .068	.0595 .0480
10	10	.248 .238	.090 .080	.0685 .0560
1/4	1/4	.325 .315	.118 .108	.0890 .0750
5/16	5/16	.357 .347	.122 .112	.0860 .0700
3/8	3/8	.427 .417	.145 .135	.1030 .0850
7/16	7/16	.498 .488	.169 .159	.1205 .1005
1/2	1/2	.568 .558	.193 .183	.1375 .1155
9/16	9/16	.638 .628	.217 .207	.1545 .1305
5/8	5/8	.708 .698	.241 .231	.1710 .1450

Table III [(a)]

Recessed Hexagon Head
Flat Pan Head
Flat Fillister Head

Screw Size	Recess and Driver Size	∅W Wing	J Recess Depth	GAGE PENET
0	0	.087 077	.033 .023	.0250 .0170
1	1	.104 .094	.039 .029	.0300 .0215
2	2	.122 .112	.046 .036	.0350 .0260
3	3	.140 .130	.053 .043	.0405 .0310
4	4	.157 .147	.059 .049	.0450 .0350
5	5	.175 .165	.066 .056	.0500 .0400
6	6	.193 .183	072 .062	.0555 .0450
8	8	.228 .218	.085 .075	.0660 .0545
10	10	.263 .253	.098 .088	.0760 .0635
1/4	1/4	.345 .335	.128 .118	.0990 .0850
5/16	5/16	.382 .372	.134 .124	.0985 .0825
3/8	3/8	.457 .447	.161 .151	.1180 .1000
7/16	7/16	.498 .488	.169 .159	.1205 .1005
1/2	1/2	.568 .558	.193 .183	.1375 .1155
9/16	9/16	.638 .628	.217 .207	.1545 .1305
5/8	5/8	.708 .698	.241 .231	.1710 .1450

(a) Gage penetration is defined as the distance to which the gage, shown on Page 5, will enter the recess.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T17:08Z
Check the source to verify that this is the current version before use.

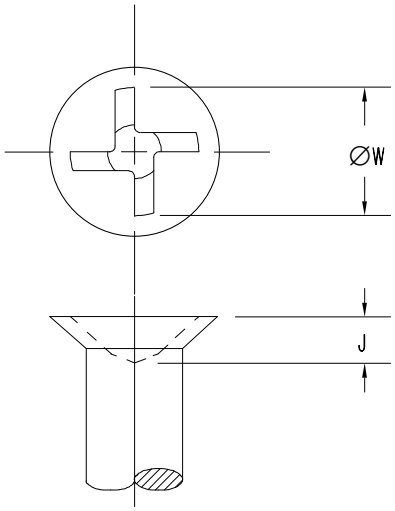


FIGURE 4

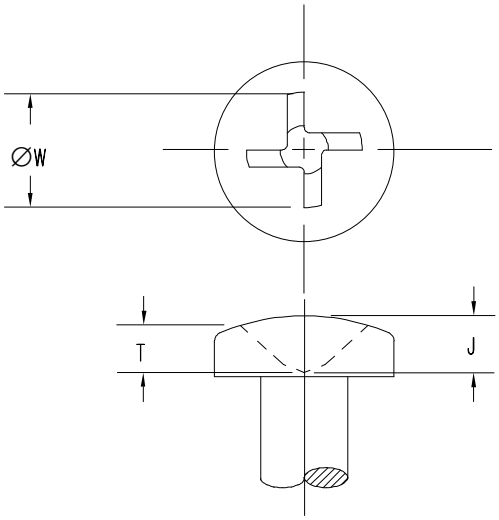


FIGURE 5

Table IV[(a)(b)]

100° Reduced Flush Head

Screw Size	Recess and Driver Size	∅W Wing	J Recess Depth	GAGE PENET
0	–	–	–	–
1	–	–	–	–
2	–	–	–	–
3	–	–	–	–
4	–	–	–	–
5	–	–	–	–
6	–	–	–	–
8	6	.182 .172	.066 .056	.0500 .0395
10	8	.215 .205	.078 .068	.0595 .0480
1/4	10	.248 .238	.090 .080	.0685 .0560
5/16	1/4	.325 .315	.118 .108	.0890 .0750
3/8	5/16	.357 .347	.122 .112	.0860 .0700
7/16	3/8	.427 .417	.145 .135	.1030 .0850
1/2	7/16	.498 .488	.169 .159	.1205 .1005
9/16	1/2	.568 .558	.193 .183	.1375 .1155
5/8	9/16	.638 .628	.217 .207	.1545 .1305

Table V [(a)]

Screw Size	Recess and Driver Size	Head				
		Pan and Brazier			Pan	Brazier
		∅W Wing	T Recess Depth	GAGE PENET	J Recess Depth	J Recess Depth
0	0	.073 063	.026 .016	.0180 .0100	.032 .022	–
1	1	.087 .077	.031 .021	.0215 .0130	.039 .029	–
2	2	.102 .092	.036 .026	.0250 .0160	.045 .035	–
3	3	.116 .106	.041 .031	.0285 .0190	.052 .042	–
4	4	.131 .121	.046 .036	.0320 .0220	.058 .048	–
5	5	.146 .136	.051 .041	.0355 .0255	.064 .054	–
6	6	.160 .150	055 .045	.0390 .0285	.070 .060	.072 .062
8	8	.190 .180	.065 .055	.0470 .0355	.083 .073	.085 .075
10	10	.219 .209	.075 .065	.0540 .0415	.096 .086	.098 .088
1/4	1/4	.286 .276	.098 .088	.0695 .0555	.126 .116	.130 .120
5/16	5/16	.357 .347	.122 .112	.0860 .0700	.153 .143	.164 .154
3/8	3/8	.427 .417	.145 .135	.1030 .0850	.181 .171	.197 .187
7/16	7/16	.498 .488	.169 .159	.1205 .1005	.209 .199	.230 .220
1/2	1/2	.568 .558	.193 .183	.1375 .1155	.236 .226	.263 .253
9/16	9/16	.638 .628	.217 .207	.1545 .1305	.270 .260	.296 .286
5/8	5/8	.708 .698	.241 .231	.1710 .1450	.303 .293	.329 .319

(a) Gage penetration is defined as the distance to which the gage, shown on Page 5, will enter the recess.
(b) See MIL - STD - 1515 for aircraft usage limitations.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T17:08Z
Check the source to verify that this is the current version before use.

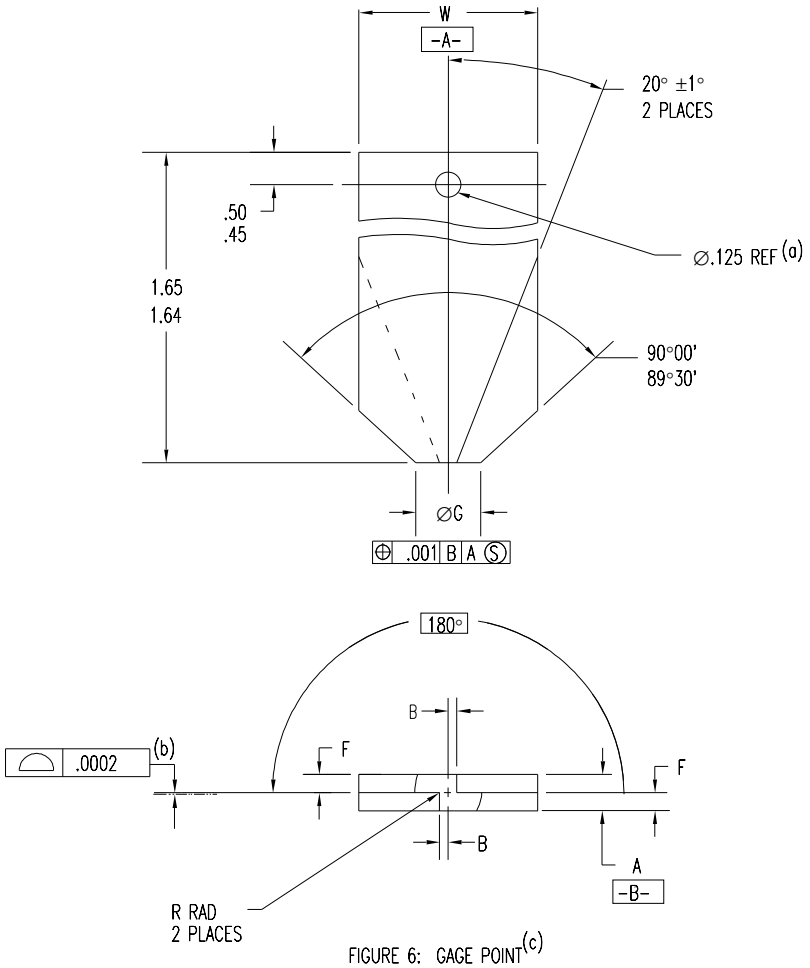


FIGURE 6: GAGE POINT(c)

Table VI. Gage Point Dimensions

Recess Size	A Ref		F +.0001		B Max	ØG ±.001		R Max	W ±.010	
	Go	Not Go	Go	Not Go		Go	Not Go		Go	Not Go
0	.0319	.0361	.0159	.0180	.004	.038	.065	.005	.250	.312
1						.045	.077			
2						.053	.090			
3	.0359	.0421	.0179	.0210		.060	.104			.375
4	.0399	.0461	.0199	.0230	.006	.068	.120	.007		
5	.0459	.0521	.0229	.0260		.076	.134			.438
6	Use MS14191									
8	Use MS14191									
10	Use MS14191									
1/4	Use MS14191									
5/16	Use MS14191									
3/8	.1459	.1521	.0729	.0760	.014	.222	.338	.030	.438	.625
7/16	.1719	.1781	.0859	.0890	.020	.258	.394	.040	.625	.750
1/2	.1959	.2041	.0979	.1020		.294	.450			
9/16	.2219	.2301	.1109	.1150		.330	.506		.688	
5/8	.2459	.2541	.1229	.1270		.367	.562			

(a) A hole of suitable size to be drilled through gage for ring.
(b) Contour requirement with unilateral tolerance (outside) applies only to centerline surface.
(c) Permanently mark manufacturer's part number and serial number.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T17:08Z
Check the source to verify that this is the current version before use.

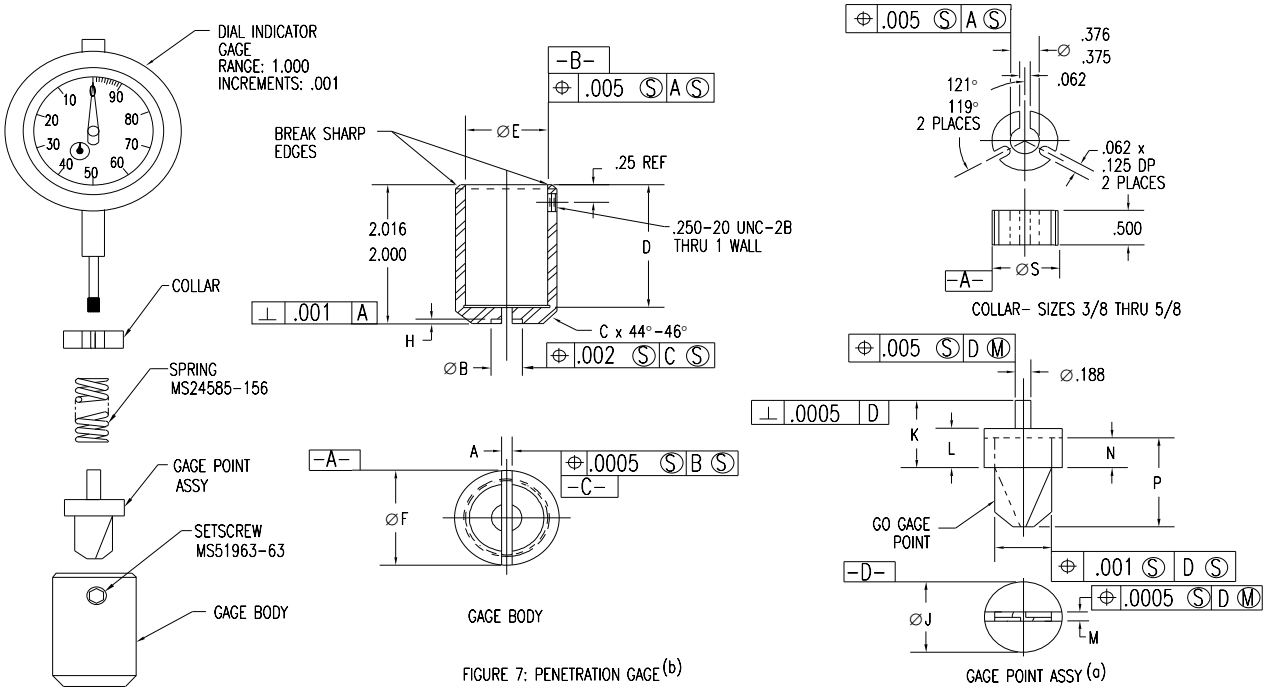


Table VII. Penetration Gage Dimensions (c)

Recess Size	A +.002	ØB +.002	C	D +.005	ØE +.001	ØF	H +.016	ØJ -.0005	K	L	M ±.001	N	P ±.005	ØS -.0005
0		.073												
1	.033	.087									.034			
2		.102	.125	1.875	.375	.875	.031	.3745	.844	.500		.188	.510	N/A
3	.037	.116									.038			
4	.041	.131									.042			
5	.047	.146									.048			
6	Use MS14191													
8	Use MS14191													
10	Use MS14191													
1/4	Use MS14191													
5/16	Use MS14191													
3/8	.147	.427	.250								.148			
7/16	.173	.498									.174			
1/2	.197	.568		1.750	.750	1.250	.062	.7495	.719	.375	.198	.312	.800	.7495
9/16	.223	.638	.125								.224			
5/8	.247	.708									.248			

(a) Component bonding at manufacturer's option.
(b) Marking shall include manufacturer's name, part number and recess size.
(c) Unless otherwise specified, tolerance: decimals ±.016.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T17:08Z
Check the source to verify that this is the current version before use.

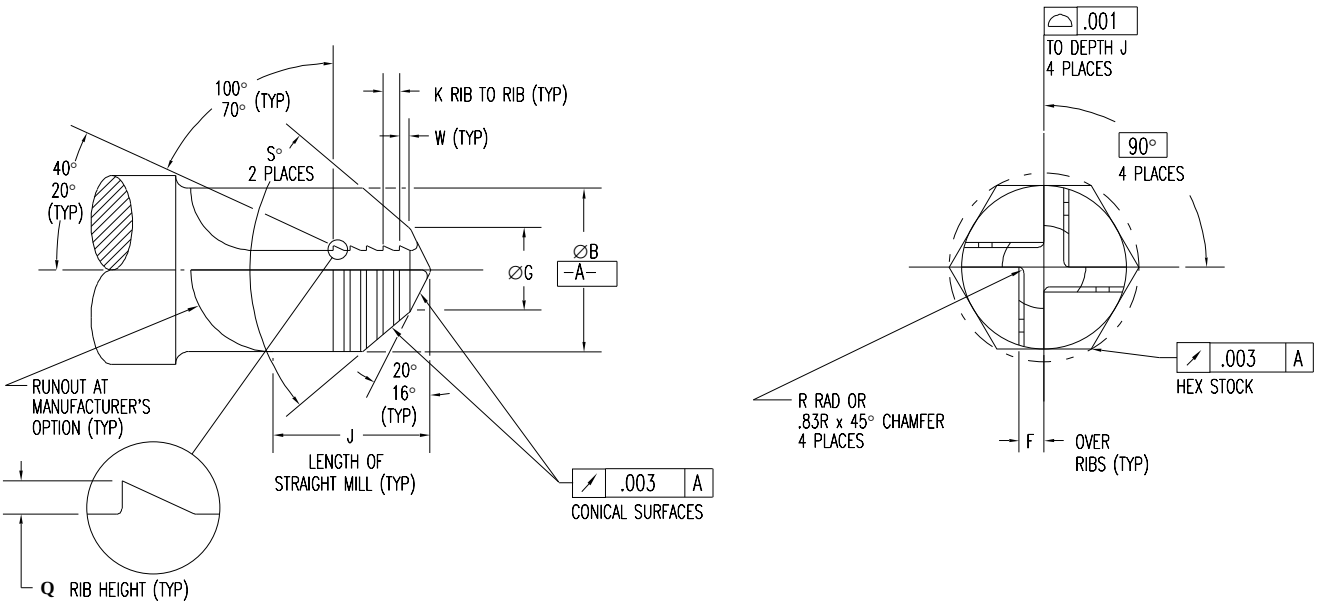


FIGURE 8: RIBBED DRIVER(a)(b)(c

Table VIII. Ribbed Driver Dimensions

Driver Size	ØB -.005	F Over Ribs Max	F Over Ribs Min	ØG -.002	J Min	K +.007	Q	R Max	R Min	S	W Max	Min Ribs
0	.185	.0155	.0140	.039	.047	.010	.003 .001	.005	.000	93° 30' 92° 30'	.009	3
1				.046								4
2				.054	.063							
3		.0175	.0160	.061							.012	
4		.0185	.0170	.069	.078			.010	.007			
5		.0215	.0200	.077								
6	Use MS14191											
8	Use MS14191											
10	Use MS14191											
1/4	Use MS14191											
5/16	Use MS14191											
3/8	.434	.0705	.0675	.223	.172	.021	.005 .002	.028	.023	91°30' 90°30'	.035	6
7/16		.0835	.0805	.259				.040	.035		.040	
1/2	.622	.0945	.0915	.295	.203			.055	.050			7
9/16		.1055	.1020	.331	.218	.026		.068	.063		.045	
5/8		.1175	.1140	.368	.250							

(a) Drivers shall have ribs on the removal wall of each wing.
(b) See MIL-B-85643 for bit procurement.
(c) The ribbed drivers in this document are covered under United States Patent 4,187,892, which expires 31 October 1997. The government does not have a royalty free license. These drivers are also produced and marketed by the Licensees of Phillips Screw Company under the designation of ACR® Muti-Ribbed Torq-Set.®

Notes:
1. Dimensions are in inches.
2. This is a design specification not to be used as a part number.
3. In the event of a conflict between the text of this document and the references cited herein, the text of this document shall take precedence.
4. Unless otherwise specified, issues of referenced documents are those in effect at time of solicitation.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T17:08Z
Check the source to verify that this is the current version before use.

Custodians:
Army - AV
Air Force - 99

Review activities:
DLA - IS
Air Force - 82

Preparing activity:
Navy - AS

Project No. 53GP-0279

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T17:08Z
Check the source to verify that this is the current version before use.